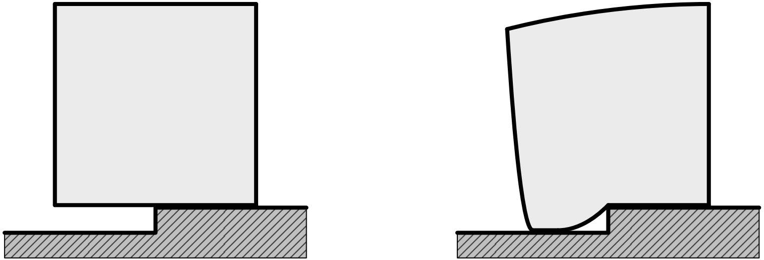
Signorini's Problem

Max Stuthmann

March 19, 2026



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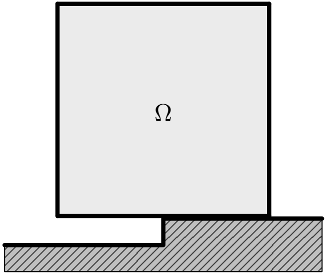
Rough Problem Setup

- *Body:* $\Omega \subset \mathbb{R}^d$, $d \in \{2, 3\}$ open, bounded, connected with Lipschitz boundary $\Gamma := \partial\Omega$.



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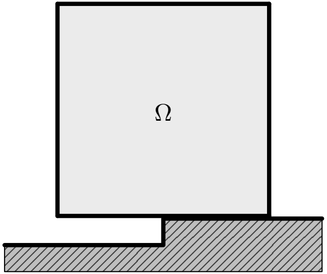

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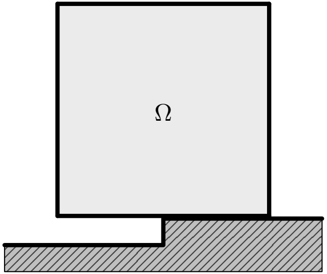

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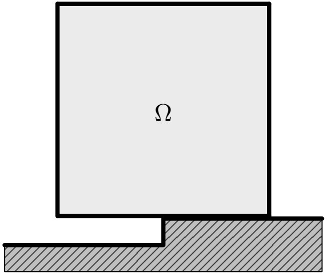

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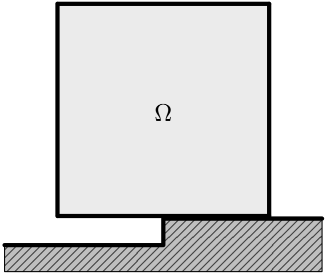
- Ω may be subject to elasticity properties and internal/external forces.
- **Not in equilibrium configuration.**



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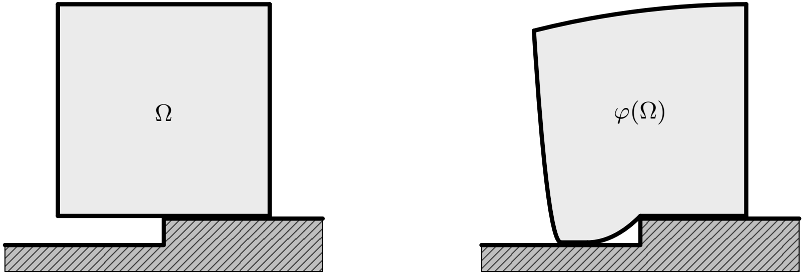

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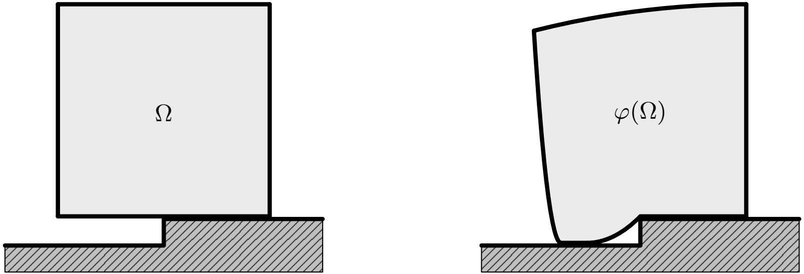
- Let $\varphi(\Omega)$ denote Ω in equilibrium configuration.
- $\varphi : \overline{\Omega} \to \mathbb{R}^d$ maps $\overline{\Omega}$ onto $\varphi(\overline{\Omega})$ and can be written as

$$\varphi(x) = x + u(x), \qquad u : \overline{\Omega} \to \mathbb{R}^d.$$



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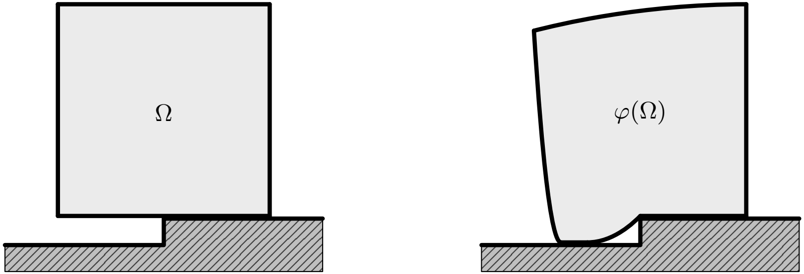
- **We aim to find the *displacement function* u.**



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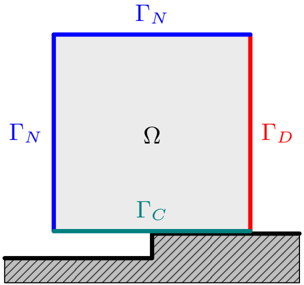
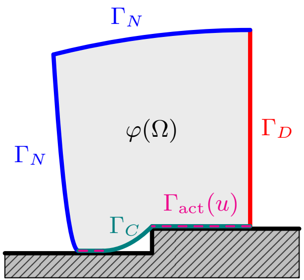
Boundary Decomposition and Preview on the Boundary Conditions

- Decompose $\Gamma = \partial\Omega$ into three relatively open and pairwise disjoint parts $\Gamma = \overline{\Gamma_D} \cup \overline{\Gamma_N} \cup \overline{\Gamma_C}$ with $|\Gamma_D|, |\Gamma_C| > 0$ and $|\Gamma_N| \geq 0$.





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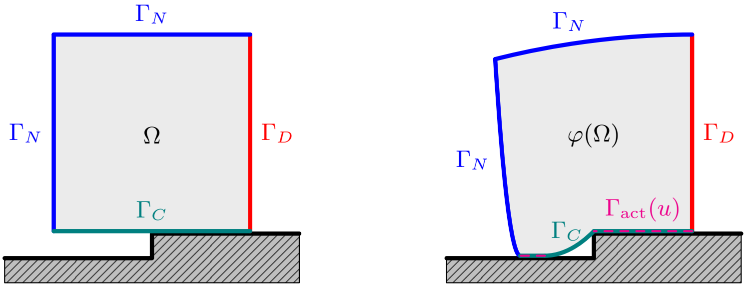

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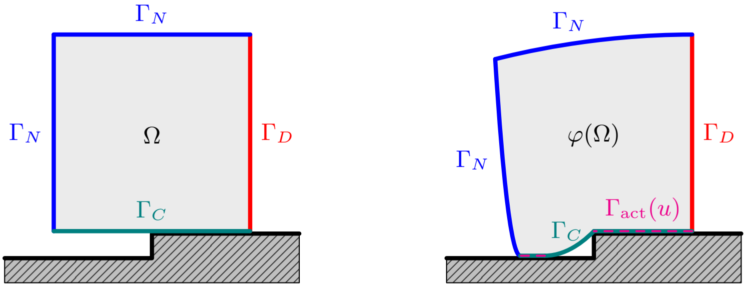

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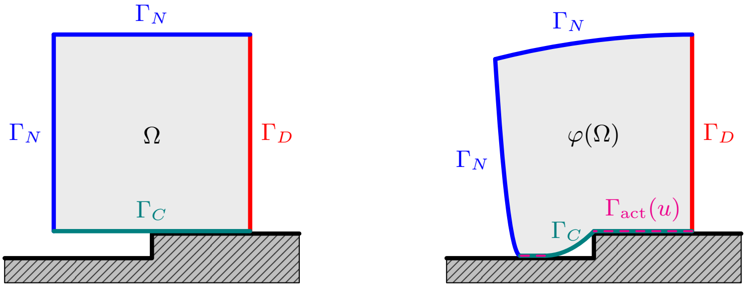

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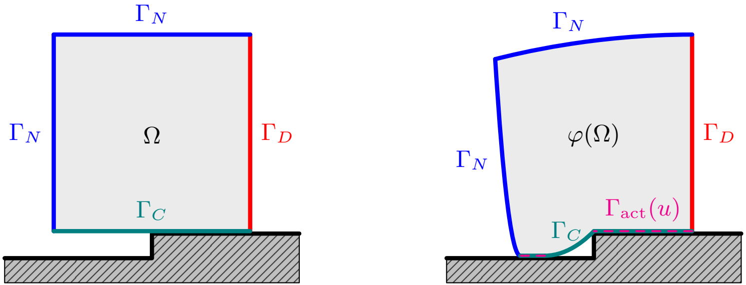

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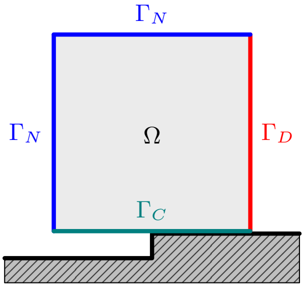
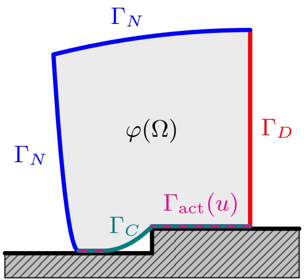
- Rigid foundation can be described by gap-function $g : \Gamma_C \to \mathbb{R}$. We assume $g = 0$.





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Basic Notations on Sobolev Spaces

- On $H^s(\Omega;\mathbb{R}^d)$ for $s = k + \theta$ with $k \in \mathbb{N}_0$ and $\theta \in [0,1)$, we use $(\cdot,\,\cdot)_{s,\Omega}$ and $\|\cdot\|_{s,\Omega}$

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- *Trace Theorem*: Restriction of smooth functions $\overline{\Omega} \to \mathbb{R}^d$ onto the boundary $\Gamma = \partial\Omega \in \mathcal{C}^{m,1}$ extends uniquely to a continuous surjective linear trace operator

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- For now, Ω is Lipschitz, i.e. $m = 0$ and we need $s = 1$, so

$$\gamma : H^1(\Omega;\mathbb{R}^d) \to H^{1/2}(\Gamma;\mathbb{R}^d)$$

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- If $\mathcal{A} : \Omega \to \mathcal{L}(\mathbb{S}^d; \mathbb{S}^d)$ is a fourth order tensor and $s \in \mathbb{S}^d$, we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x)\, s_{kl}.$$

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- $\mathcal{A}_{ijkl} \in L^\infty(\Omega)$ is sufficient for $s = 1$. Later on, we need more.

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  describes the *virtual work of internal forces*.
- *Equilibrium* between both: $a(u,v) = \ell(v) \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d)$.

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : H^1(\Omega;\mathbb{R}^d) \times H^1(\Omega;\mathbb{R}^d) \to \mathbb{R}$ *defined above is*

i) bounded, *i.e. for all* $v, w \in H^1(\Omega;\mathbb{R}^d)$ *there exists* $C_1 < \infty$ *such that*

$$|a(u,v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega}.$$

ii) $H^1(\Omega;\mathbb{R}^d)$-elliptic, *i.e. for all* $v \in H^1(\Omega;\mathbb{R}^d) \setminus \{0\}$ *there exist* $C_2 > 0$ *such that*

$$a(v,v) \geq C_2 \|v\|^2_{1,\Omega}.$$

Moreover, the linear functional ℓ *defined above is* bounded, *i.e. for all* $v \in H^1(\Omega;\mathbb{R}^d)$ *there exists* $C_3 > 0$ *such that*

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega}.$$

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- *Homogeneous Dirichlet* boundary conditions on Γ_D are incorporated in the underlying Hilbert space

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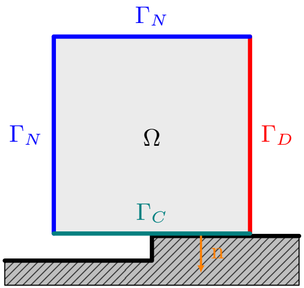

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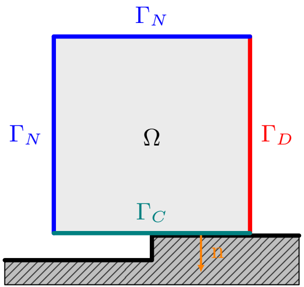

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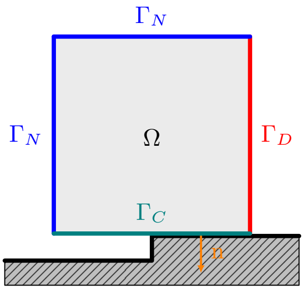
- *Nonpenetration constraint*: $u_{\mathrm{n}} \leq g = 0$ almost everywhere on Γ_C.



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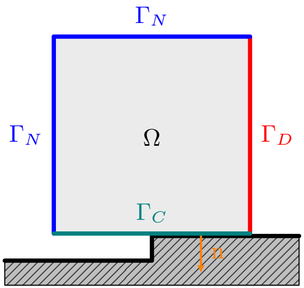

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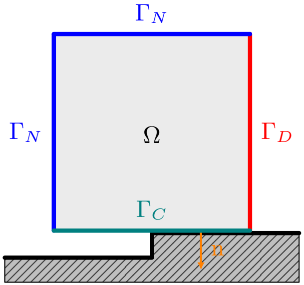

Content

Energy Minimization

- *Equilibrium*:

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Problem (Signorini's Problem – Energy Minimization)

Find $u \in K$ *such that:* $\qquad \mathcal{J}(u) = \min_{v\in K} \mathcal{J}(v).$

Well-Posedness

Proposition (Well-Posedness of Signorini's Problem)

There exists exactly one $u \in K$ that minimizes $\mathcal{J}$ over K. Moreover, the unique minimizer satisfies the a-priori bound

$$\|u\|_{1,\Omega} \leq \frac{2}{C_2} \|\ell\|_{V'}.$$

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- Stability bound follows from V-ellipticity of a w.r.t. C_2.

Content

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A function $u \in K$ minimizes $\mathcal{J}$ over K if and only if it satisfies the variational inequality

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Weak Formulations / Variational Inequality

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Content

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- Then $\operatorname{div}\sigma(u) \in L^2(\Omega;\mathbb{R}^d)$ and we can understand normal traces on $H(\operatorname{div};\Omega)$ in the following way.

Normal Traces on $H(\mathrm{div};\Omega)$

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- In the smooth case, $\gamma_\mathrm{n}(\tau)$ coincides with the classical traction $(\gamma\tau)\,\mathrm{n}$.

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- Thus, we further assume $\sigma(u) \in H(\mathrm{div}; \Omega)$ which yields $\sigma(u)\mathrm{n} := \gamma_{\mathrm{n}}(\sigma(u)) \in H^{-1/2}(\Gamma; \mathbb{R}^d)$.

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- *Useful identity*: If $-\,\mathrm{div}\,\sigma(u) = f$ almost everywhere in Ω, then

$$a(u,v) - \ell(v) = \langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma - \int_{\Gamma_N} F\cdot(\gamma v)\,\mathrm{d}s, \qquad \forall\, v \in V.$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega;\mathbb{R}^d)$ and $F \in L^2(\Gamma_N;\mathbb{R}^d)$, find $u \in K$ such that

$$
\begin{cases}
-\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, \quad (1a)\\
\langle \sigma(u)\mathrm{n}, \gamma u\rangle_\Gamma = \int_{\Gamma_N} F\cdot(\gamma u)\,\mathrm{d}s, & \quad (1b)\\
\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma \geq \int_{\Gamma_N} F\cdot(\gamma v)\,\mathrm{d}s & \forall\, v \in K. \quad (1c)
\end{cases}
$$

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\mathrm{div}; \Omega)$*. Then* $u \in K$ *solves the hybrid problem above if and only if it solves the weak problem*

$$\text{Find } u \in K \text{ such that} \qquad a(u, v - u) \geq \ell(v - u) \qquad \forall v \in K.$$

Proof: *Hybrid* $\Rightarrow$ *Weak*:

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Proof: *Hybrid* $\Rightarrow$ *Weak*:

- Subtract (1b) from (1c):

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  i.e. the weak formulation.

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- Thus, $-\operatorname{div}\sigma(u) = f$ in $\mathcal{D}'(\Omega;\mathbb{R}^d)$ and by $\sigma(u) \in H(\mathrm{div};\Omega)$, it holds in $L^2(\Omega;\mathbb{R}^d)$ as well.

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- This is exactly (1b).

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- This means

$$\langle \sigma(u)\mathrm{n},\, \gamma v\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma v\,\mathrm{d}s + \underbrace{\left[\langle \sigma(u)\mathrm{n},\, \gamma u\rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma u\,\mathrm{d}s\right]}_{=0} \qquad \forall\, v \in K.$$

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- Normal Trace: $t(u) := (\gamma \sigma(u))\, \mathrm{n} \in H^{s-3/2}(\Gamma; \mathbb{R}^d)$ is well-defined.
- If $\sigma(u) \in H(\mathrm{div}; \Omega)$, the two concepts of normal traces coincide:

$$\langle \sigma(u)\mathrm{n},\, \gamma v \rangle_\Gamma = \int_\Gamma t(u) \cdot (\gamma v)\, \mathrm{d}s \qquad \forall\, v \in V.$$

Normal and Tangential Components

- Normal and tangential splitting of $t(u)$ into

$$\sigma_{\mathrm{n}}(u) := t(u) \cdot \mathrm{n} \in H^{s-3/2}(\Gamma), \qquad \sigma_t(u) := t(u) - \sigma_{\mathrm{n}}(u)\,\mathrm{n} \in H^{s-3/2}(\Gamma; \mathbb{R}^d).$$

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- Furthermore, for $u \in H^s(\Omega; \mathbb{R}^d)$ we have $\gamma u \in H^{s-1/2}(\Gamma; \mathbb{R}^d)$ with normal/tangential splitting

$$u_{\mathrm{n}} := (\gamma u) \cdot \mathrm{n} \in H^{s-1/2}(\Gamma) \qquad \text{and} \qquad u_t := \gamma u - u_{\mathrm{n}}\,\mathrm{n} \in H^{s-1/2}(\Gamma; \mathbb{R}^d)$$

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Classical Strong Formulation under Additional Regularity

Problem (Signorini's Problem – Classical Strong Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ *and* $F \in L^2(\Gamma_N; \mathbb{R}^d)$ *find* $u \in H^s(\Omega; \mathbb{R}^d)$ *with* $s > 3/2$ *such that the* small strain elasticity equations

$$\begin{cases} -\operatorname{div} \sigma(u) = f & \text{in } \Omega, \quad (2a) \\ u = 0 & \text{on } \Gamma_D, \quad (2b) \\ t(u) = F & \text{on } \Gamma_N, \quad (2c) \end{cases}$$

and the contact conditions

$$\begin{cases} u_{\mathrm{n}} \leq 0 & \text{on } \Gamma_C, \quad (3a) \\ \sigma_{\mathrm{n}}(u) \leq 0 & \text{on } \Gamma_C, \quad (3b) \\ \sigma_{\mathrm{n}}(u)\, u_{\mathrm{n}} = 0 & \text{on } \Gamma_C, \quad (3c) \\ \sigma_t(u) = 0 & \text{on } \Gamma_C \quad (3d) \end{cases}$$

hold.

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- *Frictionless contact* $\sigma_t(u) = 0$ on Γ_C: Foundation does not exert any tangential force on the body.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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- Bulk Equation $-\operatorname{div} \sigma(u) = f$ appears in both.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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- We want to show

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Gamma t(u)\cdot(\gamma v)\,\mathrm{d}s \geq \int_{\Gamma_N} F\cdot(\gamma v)\,\mathrm{d}s \qquad \forall\, v \in K.$$

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- With $t(u) = \sigma_\mathrm{n}(u)\mathrm{n} + \sigma_t(u)$ and $\gamma v = v_\mathrm{n}\,\mathrm{n} + v_t$, we have

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- Since $v \in K$, we have $v_\mathrm{n} \leq 0$ on Γ_C and by (3b), we have $\sigma_\mathrm{n}(u) \leq 0$ on Γ_C.
- Thus $\sigma_\mathrm{n}(u)\, v_\mathrm{n} \geq 0$ on Γ_C and thus
$$\langle \sigma(u)\mathrm{n}, \gamma v \rangle_\Gamma \geq \int_{\Gamma_N} F \cdot (\gamma v)\, \mathrm{d}s \qquad \forall\, v \in K.$$

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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- Bulk equation $-\operatorname{div} \sigma(u) = f$ is contained in both problems.

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- Let $u \in H^s(\Omega; \mathbb{R}^d)$, $s > 3/2$ solve the hybrid problem.
- Bulk equation $-\operatorname{div} \sigma(u) = f$ is contained in both problems.
- Since $u \in K$, we have $u = 0$ on Γ_D and $u_\mathrm{n} \leq 0$ on Γ_C.
- It remains to prove $t(u) = F$ on Γ_N and the remaining contact conditions.

Proposition (Equivalence of Strong and Hybrid Formulation)

Let $u \in H^s(\Omega; \mathbb{R}^d)$ with $s > 3/2$. Then u solves the strong problem if and only if it solves the hybrid problem.

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- Let $g \in H^{1/2}(\Gamma_N; \mathbb{R}^d)$ such that $\mathrm{Ext}_0(g) \in H^{1/2}(\Gamma; \mathbb{R}^d)$,i.e. $g \in H^{1/2}_{00}(\Gamma_N; \mathbb{R}^d)$. Then there exists $v \in H^1(\Omega; \mathbb{R}^d)$ such that

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- Plug $\pm v$ into hybrid inequality:

$$\int_{\Gamma_N} t(u) \cdot g \, \mathrm{d}s \geq \int_{\Gamma_N} F \cdot g \, \mathrm{d}s, \qquad -\int_{\Gamma_N} t(u) \cdot g \, \mathrm{d}s \geq -\int_{\Gamma_N} F \cdot g \, \mathrm{d}s.$$

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- Therefore,

$$\int_{\Gamma_N} (t(u) - F) \cdot g \, \mathrm{d}s = 0 \qquad \forall\, g \in H^{1/2}_{00}(\Gamma_N; \mathbb{R}^d) \qquad \implies \qquad t(u) = F \text{ on } \Gamma_N.$$

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$\square$

Content

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- Λ_C encodes the weak condition that the normal contact force can only be compressive.

Things that did not fit in this Presentation – Mixed Formulation

- We had $\sigma(u)\mathrm{n} = \gamma_{\mathrm{n}}(\sigma(u)) \in H^{-1/2}(\Gamma; \mathbb{R}^d)$ for $\sigma(u) \in H(\mathrm{div}; \Omega)$.
- Similarly, one defines *local* normal traces

$$\sigma_C(u)\mathrm{n} := \gamma_{\mathrm{n},C}(\sigma(u)) \in (H_{00}^{1/2}(\Gamma_C; \mathbb{R}^d))'.$$

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Problem (Signorini's Problem – Mixed Weak Formulation)

Let $|\Gamma_N| = 0$. *For given* $f \in L^2(\Omega; \mathbb{R}^d)$ *find* $(u, \lambda) \in V \times \Lambda_C$ *such that*

$$\begin{cases} a(u,v) - \langle \lambda,\, v_{\mathrm{n}}\rangle_{\Gamma_C} = \ell(v) & \forall\, v \in V, \quad (5a) \\ \langle \mu - \lambda,\, u_{\mathrm{n}}\rangle_{\Gamma_C} \geq 0 & \forall\, \mu \in \Lambda_C. \quad (5b) \end{cases}$$

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- Strong Formulation: smooth
- Variational and mixed formulations are natural starting points for finite element methods.

Thank you!

Questions?

Max Stuthmann